

ChE 322 Thermodynamics Fall 23 8 12 23 version

Prof: Keith Johnston, Office Hours: Zoom M and W 4:30- 5:30 or by appointment.

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Prerequisites: ChE 210, 317, CH 353, and M 427J with at least C- in each

Lecture: TTh 11 to 12:30 GLT 1.106

Recitation: Wed 1 to 2 CPE 2.220 (one hour total)

Goals: The objective of this course is to gain a solid foundation of the laws and concepts and language of thermodynamics for pure materials and mixtures and to apply the concepts to solving problems of interest in chemical processes and in other fields.

Textbook: Koretsky, Milo , Engineering and Chemical Thermo., Wiley 2nd edition.
University Coop: Either hard copy or digital rental This book is fantastic as it is written by a leading educator!

Canvas: We will use files, announcements, assignments, and quizzes.

Evaluation: Students are strongly encouraged to comment on the course and instructor online.

<u>Week</u>	<u>Topic (Koretsky)</u>
1-2	Work, energy Balance: steady-state and unsteady state (Chpt 2)
2-4	Entropy and second law, Power cycles and refrigeration (Chpt 3),
5	Equations of State(Chpt 4)
6-7	Thermodynamic Properties, Residual properties (Chpt 5)
8-9	Phase equilibria I (Chpt 6)
10-11	Phase equilibria II: Fugacity (Chpt 7)
12-13	Phase equilibria III: VLE, LLE, VLLE, SLE (Chpt 8)
14-15	Chemical equilibria (Chpt. 9)

Grading method: 3 Hourly exams (60%), Final (30%), Homework and quizzes (10%)

Exams: dates to be determined

Closed book. The summary equation sheet may be used. You may also use one page of your own notes front side only. Please contact me prior to the exam if you can not attend and Ill do my best to accommodate.

Homework and recitation quizzes: Homework will be assigned in canvas for most classes to encourage continuous learning. Upload scanned copy in assignments in canvas.

Homework and examples: Canvas table will explain homework by type and difficulty. The goal is to help you to classify types of problems, as this will help you prepare for tests. A guide will be updated for how to study each chapter.

Coordination of primary lecture notes, supplemental notes, learnchemE screencasts and text: Pls see files on Canvas. The notes are not meant to be self sufficient, but are instead intimately tied to the text.

LearnChemE screencasts: <http://www.learncheme.com>. See **thermodynamics**. All of you learn differently. We wish to reach all students from top to bottom of background. These highly acclaimed optional screencasts for PCs to tablets to phones may be very helpful to many of you. Canvas table will further elaborate.

Knowledge, Abilities, and Skills Students Before Entering:

1. Units, material and energy balances, the use of steam tables and P-H charts (ChE 317).
2. The first law of thermodynamics, enthalpy, and heat capacity (ChE 317).
3. Ideal gas and real fluid behavior (ChE 317).
4. The first and second laws of thermodynamics (Ch 353).
5. State and path-dependent functions in the solution of chemical problems (Ch 353).
6. Solution thermodynamics and chemical reaction equilibria (Ch 353).

Knowledge, Abilities, and Skills Students Should Gain:

1. The student should be able to apply energy and entropy balances to open and closed systems and to evaluate the thermodynamic efficiency of compressors, turbines, Rankine cycles and refrigeration cycles.
2. The students should be able to understand and calculate property relationships for pure materials and mixtures by using the calculus of state functions, equations of state, tables, figures, and empirical activity coefficient models.
3. The student should be able to understand the concepts and to solve phase equilibria problems involving vapor, liquid and solid phases for ideal and nonideal systems.
4. The student should be able to understand the concepts and to solve problems in homogeneous and heterogeneous reaction equilibria, by extending freshman chemistry concepts to include nonideal behavior.

Impact on Subsequent Courses in Curriculum:

Thermodynamic properties, phase equilibria and chemical reaction equilibria play an important role throughout chemical engineering, e.g. in ChE 360, ChE 363, ChE 372 and ChE 473K. Thermodynamics is one of the main pillars of chemical engineering; others include transport phenomena and reaction kinetics.

Mental wellness comments during pandemic or otherwise:

The professor, TA and fellow students will look out for the mental wellness of the students. Uplifting news, stories and kindness and empathy/compassion are greatly appreciated.



UT counseling and mental health center: great resource <https://cmhc.utexas.edu/>

Respect and civility: In this course, it is expected that you will hold yourself and your classmates to the values of respect, diversity, curiosity, and inclusion when you speak and also when you listen.

Academic Integrity: Each student in the course is expected to abide by the University of Texas Honor Code: “As a student of The University of Texas at Austin, I shall abide by the core values of the University and uphold academic integrity.” Plagiarism is taken very seriously at UT. Therefore, if you use words or ideas that are not your own (or that you have used in previous class), you must cite your sources. Otherwise you will be guilty of plagiarism and subject to academic disciplinary action, including failure of the course. You are responsible for understanding UT’s Standards of Conduct and Academic Integrity principles.

Tips to maximize success:

1. It is an honor to have you in our department and in my class as you have already excelled. Continue to develop a passion for knowledge and life-long learning.
2. Review info where you first learned it. Reread key sections in calculus book, P Chem. book. The more you review the better you will retain it.
3. See us in office hours. Participate in lecture and recitation. Students often chat with me right after class as well.

4. Find other students that study together if you can.
5. **It will take~ 8 hours/week outside of class to read the book including examples in chapters (in conjunction with notes), learn the theory and concepts, and to do the homework problems.**
6. Strive to understand the theory and concepts needed to solve the problems rather than just finding an equation to solve a specific problem. Study examples in the book chapters recommended in Canvas.
7. Learn to identify classes of problems based on theoretical concepts. It takes practice and time.
8. Maximize what you get out of the lectures by keeping up and understanding all concepts after each lecture. If you get behind, try to immediately catch up to max. efficiency for learning.
9. Look ahead in the book for about 5-10 min before each lecture. The brain will then be activated and eager to learn very efficiently and deeply.
10. Avoid the short cut to use old homework solutions as a crutch. It is far easier to follow somebody else's solution than to construct your own. See if you can find a related example in book or notes.
11. Students should do all homework problems and study solutions after turning in homework as needed on Canvas. Save hws when you get them back so you can study your approach vs. posted solutions.
12. Always check units in eqns. They are important. Check with factor label. This approach helps with exam grades.